Final Report

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Part IIA Project: *Neural Control with Adaptive State Estimation* (GG4)
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Abstract

We are provided a simulation of a prosthetic hand controlled through a fixed Brain–Machine Interface (BMI) driven by a black-box Linear Gaussian State–Space Model (LGSSM) `Brain`, where only 16-dimensional `Brain` observations and cartesian hand position are available as feedback. We identify system matrices using Eigensystem Realisation Algorithm (ERA) and Expectation-Maximisation (EM) with latent dimension $n_x = 4$, highlighting benefits over Canonical Variate Analysis (CVA) and EM. We demonstrate control over population mean frequency spectrum, demonstrating the strengths of Linear Quadratic Integral (LQI) over Linear Quadratic Gaussian (LQG) and Model Predictive Control (MPC) in Root Mean Square (RMS) error. We develop a pipeline to control the hand via 2 antagonistic muscle pairs which act on the shoulder and elbow joints that involves frequency band indentification, gain matrix estimation, and a sequential damped strategy which mitigates isses of cross-talk. We achieve a final distance of 3.9 ± 2.76 with a time to threshold of 10 cm of 308 ± 155 steps, with a 100% threshold reach rate. We find that the system's linearity permits a rhythm based BMI run open-loop.

Introduction

The system is a arm driven from neural signals, where the controller sees neither brain state nor muscle commands. A black box LGSSM, irreducible measurement noise, and a fixed nonlinear BMI decoder sitting in the loop provide challenges to control. This project explores how we can estimate system parameters and achieve spectral control. In addition, we compose a control strategy and explore its evaluation, sensitivity to its parameters, robustness to noise, limitations, and strengths.

Problem statement

We are given a `Brain` and a BMI, which controls an arm. We are tasked with controlling the position of the hand at the end of the arm. The `Brain` takes $n_u = 2$ inputs and produces $n_y = 16$ outputs; we assume it is a LGSSM. The BMI takes the 16-dimensional observations from the `Brain` and applies a fixed linear decoder to map them to 2 latent states: x_1, the muscle selection rhythm; and x_2, power rhythm. The BMI stores a rolling history of x_1, which is passed through 4 fixed band–pass filters which measure how strongly x_1 oscillates at that muscle's defined excitation frequency. A history of x_2 is also stored in state, and the current value is compared to a half– and full–period in the past: if strong oscillation is found, evidence accumulates; otherwise, evidence decays, producing a smooth power state. Muscle activation is the product of activation and the selection vector. Each muscle produces force proportional to its activation, and for each joint, the net force is the sum of signed muscle forces. Positive forces represent flexion; negative forces represent extension. A dry-friction threshold must be overcome, and once this is overcome, the remaining net force (once subtracting the dry friction) is divided by wet friction representing viscous resistance. Joint angles are integrated, and 2-link arm kinematics give the hand position.

Rules of engagement

The holistic view of this project is to be able to reason and interpret a controller which takes observations from a brain (e.g. via electrodes on the scalp) to move a prosthetic arm. Critically, the brain is a black box and its model "parameters" cannot be directly observed. In the spirit of this view, we propose specific rules of engagement surrounding our analysis. We do not attempt to decompile the provided `Brain` executables to derive the system matrices. We do not average over multiple `Brain` seeds, removing noise to derive system matrices, we reserve this for evaluation purposes only to demonstrate robustness against noise. Where we do run experiments that are not physically realisable, we do this to demonstrate the direction of our analysis, and we replicate results in empirically valid ways alongside. We do not directly use the weights of the BMI in our control efforts: we allow ourselves to read their values for experimental design, but any results we must prove empirically before use. We only allow the use of the 16-dimensional observations from the `Brain` and the hand position given by the BMI for closed-loop control; we do not rely on the internal state of the BMI (e.g., arm angles or latent state variables) for feedback. We also do not expect or attempt to control the elbow joint past 180° to mimic the capabilities of an anatomically accurate arm. We motivate these constraints as to achieve resilience and independence to both: inter-individual variation in brains (and therefore system matrices); and manufacturers of prosthetic arms (and therefore BMI decoding, state, and arm kinematics defined by model weights). Finally, during control, the hand must be kept in the target position until the end of the trial.

System identification

As we assume the `Brain` is a LGSSM, we explore system identification as a means to understand the system matrices defining the model:

$$\begin{aligned} \underline{x}_{t+1} &= \underline{A}\,\underline{x}_t + \underline{B}\,\underline{u}_t + \underline{w}_t, & \underline{w}_t &\sim \mathcal{N}(\underline{0}, \underline{Q}) \\ \underline{y}_t &= \underline{C}\,\underline{x}_t + \underline{o}_t, & \underline{o}_t &\sim \mathcal{N}(\underline{0}, \underline{R}) \end{aligned}$$

Identification evaluation

To evaluate our estimates, we have two strategies: the reconstruction of observations and the reconstruction of frequency response. To measure the reconstruction of observations, we use the coefficient of determination defined with the Frobenius norm:

$$R^2 = 1 - \frac{\left\| Y - \widehat{Y} \right\|_F^2}{\left\| Y - \overline{Y} \right\|_F^2}$$

This form is chosen for its effectiveness at aggregating errors across all time points and all observation dimensions. We weight this up against its disadvantages, which include ignorance of observation covariance and temporal correlations. To measure the reconstruction of frequency response, we inspect a frequency map derived analytically from estimates of system matrices, and compare characteristics with empirical results from a a single run of the `Brain`. We find the transfer function of the LGSSM, dependent on only the deterministic components (matrices $\underline{A}$, $\underline{B}$, and $\underline{C}$):

$$\underline{H}(z) = \underline{C}\,(z\underline{I} - \underline{A})^{-1}\,\underline{B}$$

We define an estimate to reconstruct the frequency response well if it can minimise the error between analytic and empirical values of $|H(e^{j\omega})|$ across all neurones and frequencies $[0, \pi]$. Given the emphasis on rhythm in the BMI we weigh frequency response reconstruction more than observation reconstruction if a compromise is necessary.

Identification methods

We run two `Brain` instances with and without input to isolate and subtract the noise. We can then calculate the exact Markov parameters (impulse-response coefficients of the LGSSM $h_k = \underline{C}\underline{A}^{k-1}\underline{B}$), construct Hankel matrices (block matrix built by stacking time-shifted Markov parameters), and form a singular-value spectrum. With this analysis we find σ_i/σ_1 fall after $i = 6$, indicating the true latent dimension number to be $n_x = 6$ (Figure 1). However, this method is not physically realisable: you cannot run the same physiological brain twice with the same random state. This would mean rewinding a brain, re-run a trial without a stimulus, and subtract: noise is truly stochastic outside of a simulation. A physically realisable single-run Ordinary Least Squares (OLS) regression (fitting system matrices by regressing each step on the prevous one) estimate uncovers a noise floor that obscures modes 3–6. Balancing likelihood and model complexity with Bayesian Information Criterion (BIC) model selection, we identify $n_x = 2 \neq 6$ (Figure 1).

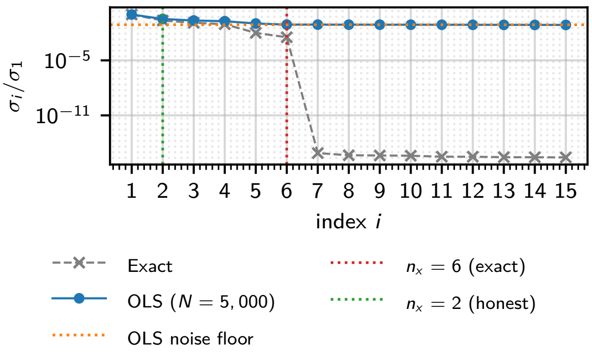

Figure 1: Normalised Hankel singular values σ_i/σ_1 of the 40×60 block Hankel matrix, comparing the exact same-seed-subtraction spectrum (a theoretical reference for the true order) with the physically realisable single-run OLS spectrum. Source

We plot a heatmap of $|H(e^{j\omega})|$ and find that CVA and EM to reconstruct u_1 well, but not u_2 over all n_x candidates $\{2, 4, 6\}$ (Figure 4, N.B. only $n_x = 4$ plotted). CVA is a subspace method that defines the state as the linear combination of past data most correlated with future data. EM is a iterative maximum-likelihood estimator, where the E-step runs a Kalman smoother to infer states, and M-step re-estimates the system matrices assuming them to be true. We turn to an estimation method that uses the ERA (a method which derives deterministic system matrices $\underline{A}$, $\underline{A}$, and $\underline{A}$ directly from the singular-value spectrum of the Hankel matrix of Markov parameters) and EM that utilises OLS-estimated Markov parameters and find similar results for $n_x = 2$: good reconstruction of u_1 but poor for u_2. However, in contrast to CVA+EM, reconstruction of the u_2 frequency response improved with $n_x = 4$ and $n_x = 6$ (Figure 4). More generally, we observe that the `Brain` acts as a low-pass filter. The observation reconstruction quality remained similar between CVA+EM and ERA+EM (Figure 2).

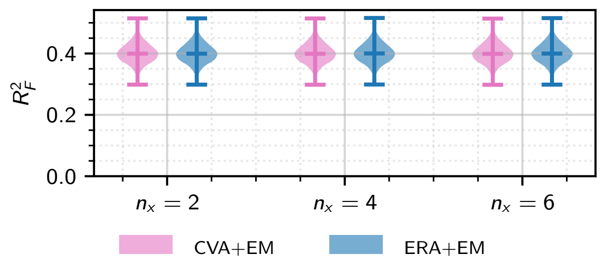

Figure 2: Frobenius R_F^2 of one-step-ahead Kalman predictions across $N = 1000$ held-out trials of $T = 2000$ steps (first 100 steps excluded as Kalman transient). Every estimator and n_x combination clusters at a median $R_F^2 \approx 0.40$. Source

As there was not a huge difference between $n_x = 4$ and $n_x = 6$, we select ERA+EM and $n_x = 4$, which yields system matrices visualised in Figure 3.

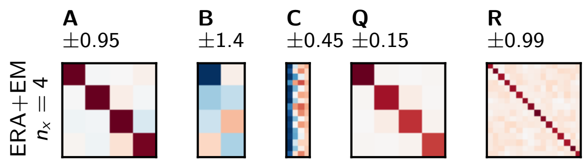

Figure 3: Identified system matrices A, B, C, Q, R of the chosen model. Diverging colour scale is symmetric about zero. Source

Closed-loop spectral control of brain observations

We test whether closed-loop stimulation can reproduce a prescribed frequency spectrum in neural population mean activity. We motivate the following evaluation framework and method with the emphasis on controlling rhythms in the `Brain` observations for influence over BMI latent variables and the arm downstream.

Spectral control evaluation

We use the following metrics to evaluate each controller: spectral RMS error, which measures amplitude deviation at target frequencies; the maximum amplitude error, the worst-case per-component deviation; and the off-target power ratio, the mean spectral amplitude at non-target frequencies, relative to mean target amplitude.

Spectral control method

We offer 3 candidates for control strategies: LQG to respect the `Brain`'s LGSSM structure; LQI to correct steady-state error; and MPC to optimise for and appreciate constraints on $u_i \in [0, 1]$. LQG involves a Kalman filter feeding a Linear Quadratic Regulator (LQR) feedback gain that aims to minimise a quadratic cost on state error and control effort, a controller specific to the `Brain`'s LGSSM structure. We augment this with an integrator on the latent state tracking error, which attempts to drive steady-state error to 0. MPC is a horizon-based controller that re-solves a constrained finite-horizon optimisation at each step, taking the optimal first value each time. This optimisation step allows it to respect constraints on inputs.

Spectral control results

Inspecting the raw population mean traces, we find LQG overestimates the population mean with a steady-state error, which is removed effectively by LQI. MPC fails to reconstruct the higher frequency components. We select LQI due to its lower control effort compared to LQG and better reconstruction performance over MPC.

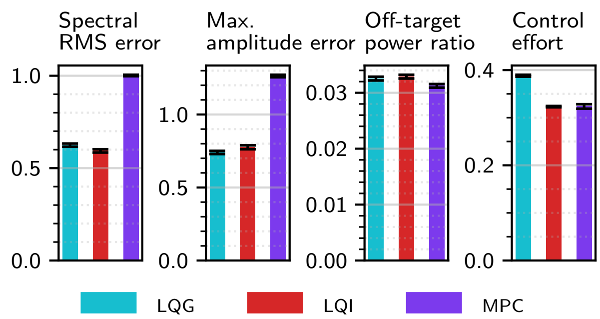

Figure 6: Spectral tracking metrics across $N = 20$ independent Brain seeds, for controllers tuned by random search (8 trials, seed 1) on the target of Figure 5. Bars show the mean; error bars the standard error. Panels left to right: spectral RMS error, maximum amplitude error, off-target power ratio. Source

1 Hand control

The mechanism of the BMI of muscle selection and a shared power state means that simultaneous control of different muscles is unlikely, as there is no granular control to direct different muscles at different power at the same time. This constrains our control strategy to be a sequential one, directing one muscle at a time.

Inverse kinematics

Inverse kinematics is used to map cartesian hand positions to shoulder and elbow angles. With equal link lengths $L = 30$, the law of cosines fixes the elbow bend, and the isosceles geometry places the shoulder a half-bend below the bearing to the target:

$$\theta_e = \arccos \frac{r^2 - 2L^2}{2L^2},$$
$$\theta_s = \operatorname{atan2}(y, x) - \tfrac{1}{2}\theta_e$$

Any elbow motion therefore shifts the shoulder's required offset, the kinematic origin of the cross-talk.

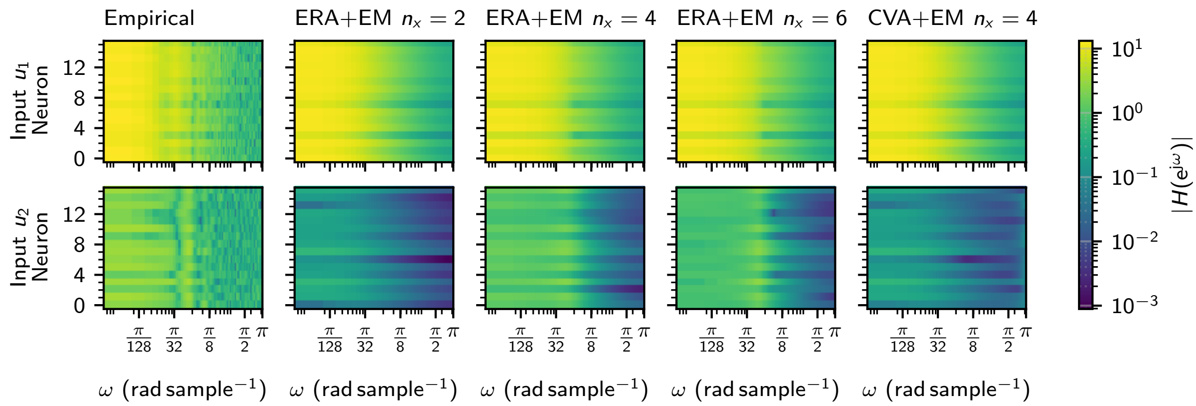

Figure 4: Per-neuron frequency selectivity $|H(e^{j\omega})|$ for inputs u_1 (top row) and u_2 (bottom row), all neurons against $\omega \in [0, \pi]$ on a shared logarithmic colour scale. Leftmost column: empirical response (single-run OLS, 5000 steps, DFT zero-padded to 512 points). Remaining columns: analytic ERA+EM responses for $n_x \in \{2, 4, 6\}$ and CVA+EM at $n_x = 4$ for comparison. Source

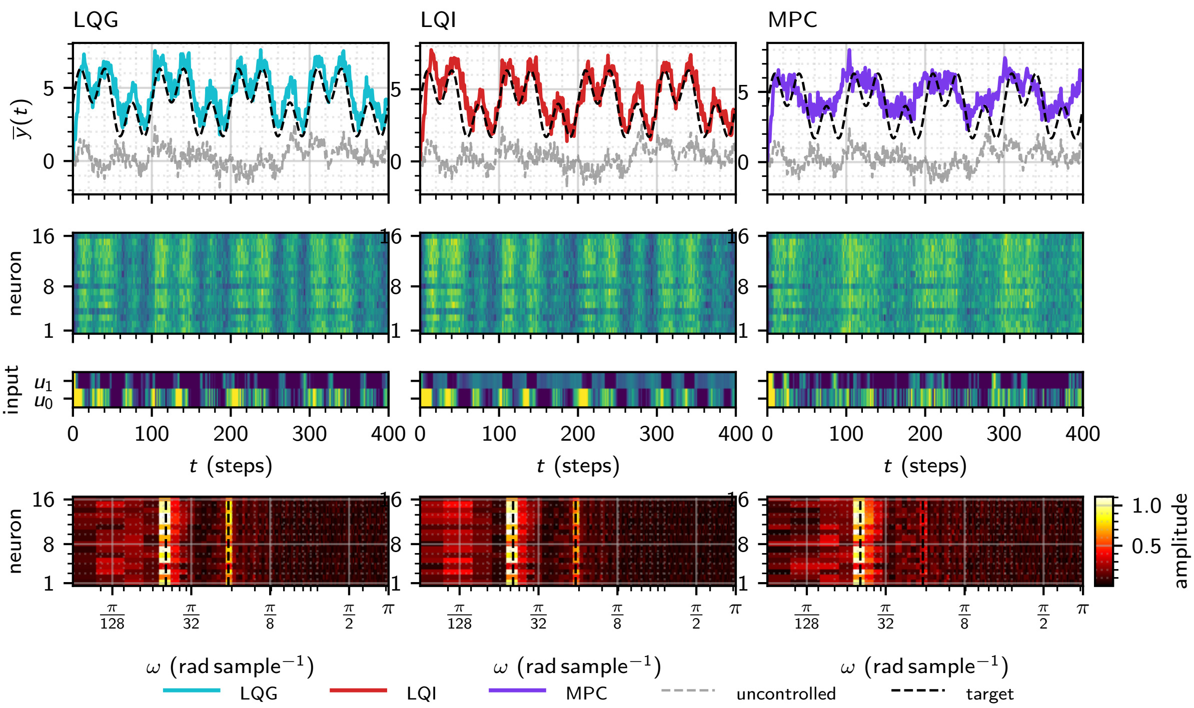

Figure 5: Closed-loop spectral control of the population mean. Target: two sinusoids at $f \in \{0.01, 0.03\}$ cycles per sample (amplitudes $A \in \{1.50, 1.50\}$) about a mean of 4; each controller tuned by random search (8 trials). Columns are LQG, LQI, MPC. Rows, top to bottom: population mean $\overline{y}(t)$ with the uncontrolled baseline and target reference; per-neuron observations $y(t)$; applied inputs $u(t)$; per-neuron amplitude spectra with target frequencies marked. Source

The sign of θ_e is the elbow-up/elbow-down redundancy; as only the hand position is observable, we resolve it toward the previous pose, keeping the inferred joint trajectory continuous and avoiding spurious elbow flips. Targets beyond the reachable disk ($r > 2L$) are clipped to the nearest pose.

Evaluating hand control

We aim to minimise the final distance to target, the Euclidean distance between the hand and the target at the end of a trial. We define a success threshold at 10. We aim to maximise the reach rate, the fraction of trials that reach within this success threshold. This metric captures reliability across diverse targets in the reachable area: a controller with low mean final distance distance but occasional failures is not practically useful for a prosthetic. We minimise the time to threshold, the number of steps until the hand first enters the distance success threshold: a useful prosthetic should converge in reasonable time to provide utility. We maximise the sector fraction, the proportion of time the hand spends in the optimal sector: an angular wedge containing the target and the start position. We minimise the rotational distance, the total angle swept by the hand trajectory over the trial. These two metrics penalise inefficient trajectories (e.g. looping behaviours), which would be inconvenient and unpredictable for a prosthetic. We minimise the control effort, the sum of square brain inputs, in an effort to make inputs more interpretable and efficient.

Muscle selection

We sweep frequencies for both inputs and measure shoulder and elbow angles, determined by inverse kinematics to find the specific frequencies that excite specific muscles. We empirically recover the frequency bands defined in the BMI weights using a single seed, shown in Table 1 and Figure 7.

Muscle	ω^*_{true} (rad step^{-1})	ω^*_{rec} (rad step^{-1})	$\Delta\omega$ (rad step^{-1})
S+	0.4378	0.4263	−0.0115
S−	0.7854	0.7800	−0.0054
E+	1.2875	1.3006	0.0131
E−	1.9764	1.9458	−0.0306

Table 1: True bandpass-filter centre frequencies ω^*_{true} (from the BMI weights) against the values ω^*_{rec} recovered by the open-loop sweep (Figure 7); $\Delta\omega = \omega^*_{\text{rec}} - \omega^*_{\text{true}}$. Rows are ordered by ascending frequency: shoulder flexion (S+), shoulder extension (S−), elbow flexion (E+), elbow extension (E−). Source

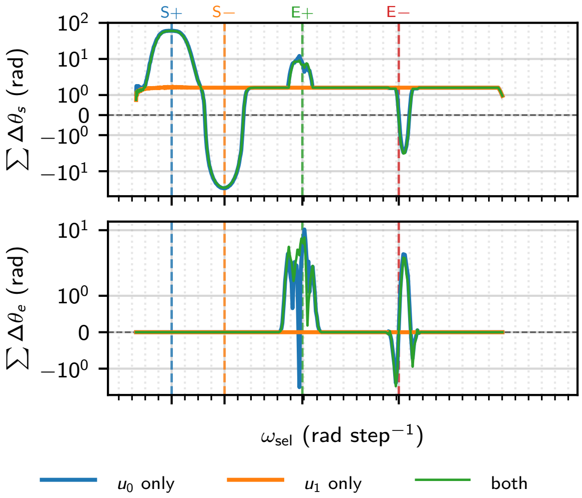

Figure 7: Open-loop sinusoidal drive (amplitude $A = 0.45$ about 0.5) applied to u_0 or u_1 independently at each of 512 log-spaced frequencies $\omega_{\text{sel}} \in [0.188, 2.639]$ rad step^{-1}. Top: signed cumulative shoulder displacement $\sum \Delta\theta_s$; bottom: cumulative elbow displacement $\sum \Delta\theta_e$, each summed over a 900-step post-warmup window (symlog, linear threshold ± 1 rad; 5-sample moving average). Vertical lines mark the four bands, located on the u_0 response by peak displacement (S$\pm$) and peak elbow selectivity $|\Delta\theta_e|/(|\Delta\theta_e| + |\Delta\theta_s|)$ (E$\pm$). Source

We observe that amplitude decreases with frequency, making it harder to drive muscles that require higher frequencies. This may be due to the low-pass action of the `Brain`. We also observe cross-talk for elbow movements but not shoulder movements: the shoulder can be moved independently of the elbow, but the elbow cannot be moved indepedently of the shoulder.

We then ask whether neural observations carry a muscle-specific spatial code: if different neurons respond preferentially at each selection frequency, the control target could be weighted to favour one muscle over another. For each selection frequency we record the full $n_y = 16$ D amplitude spectrum and normalise the per-neuron profile by its own neuron-mean (Figure 8).

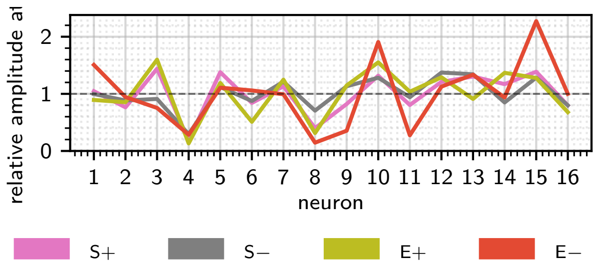

Figure 8: Neural correlates of muscle selection. For each of the four selection frequencies identified in Figure 7, a sustained open-loop sinusoidal trial ($u_0 = 0.5 + 0.4\sin(2\pi f^* t)$, $u_1 = 0.5$) recorded $n_y = 16\,\mathrm{D}$ neural observations over 1024 steps following 500 burn-in steps. Each line shows the amplitude at ω^*_{sel} per neuron, normalised by the muscle's own neuron-mean; the dashed line marks the mean ($= 1$). Profiles that collapse onto each other confirm that every muscle recruits the same relative neural activation pattern, so the $16\,\mathrm{D} \to x_1$ projection is not muscle-specific. Source

Muscle identity is therefore encoded temporally by the oscillation frequency of x_1, not spatially by which neurons are active (Figure 4). No differential weighting of neurones in the spectral control target offers muscle selectivity beyond what the BMI's bandpass filters already provide. This justifies controlling the population mean at the spectral target.

Control method

The core pipeline is a 3 step process. First is band-identification, an open-loop sinusoidal sweep over both brain input channels, with cumulative shoulder and elbow displacement (inferred via inverse kinematics) recorded per frequency. The four bands are identified, which correspond to S+, S−, E+, and E−. The next step is gain calibration, where a short-closed loop LQI trial at each band frequency targets a sinusoidal spectral setpoint, and mean joint angle per unit amplitude forms a 2×4 gain matrix where rows are joints, and columns are bands. The third is the main control loop, where inverse kinematics converts cartesian target to a joint-angle goals, band amplitudes are solved from the gain matrix and an LQI controller optionally closes the loop on the population mean.

The elbow to shoulder cross-talk motivates a control strategy that first achieves the correct elbow angle, then reduces control gain over the elbow, and focuses on using S+ and S− to correct the shoulder angle. In addition, during the initial correction angle phase, we should alternate between correcting the elbow angle and correcting the shoulder angle, as attempting to move the elbow moves the shoulder. To mitigate cross-talk further, we drive the shoulder in the opposite direction of the expected cross-talk.

We always begin with targeting elbow error first. Each elbow/shoulder phase has a minimum length and maximum length. In addition, when controlling the elbow, there is a maximum shoulder drift, such that if upon moving the elbow the shoulder drifts too much, we switch to controlling the shoulder. We also provide per-band gain multipliers applied to the gain matrix values before the solve, to further tune individual band aggressiveness. Once the elbow angle is below threshold, we begin damping the shoulder velocity in an effort to reduce oscillations.

Hyperparameter optimisation

With so many model parameters defined by the complex sequential controller design, we turn to hyperparameter optimisation to search the parameter space for an optimal solution. We use Tree-structured Parzen Estimator (TPE) via the `optuna` library to run two phases: first, a large set of Bayesian Optimisation (BO) trials; then the top trials are re-evaluted on a full target grid, and the best is selected. BO is effective as it samples where improvement is expected; its sample efficiency is more feasible with expensive trials over grid search.

We define an objective score:

$$\mathcal{S} = \overline{d}_{\text{final}} - 20\,\overline{f}_{\text{sector}} - 10\,\overline{p}_{\text{reached}} + 50\,\overline{\max(0,\ \theta_{\text{rot}} - 2\pi)}$$

This directly penalises final distance, inefficient trajectories, and rewards successful reach rate to create a more interpretable optimisation process. Figure 9 shows how the optimisation process did not reduce final distance or the time to threshold, but instead made the trajectory more sensible, increasing the sector fraction and reducing the rotation.

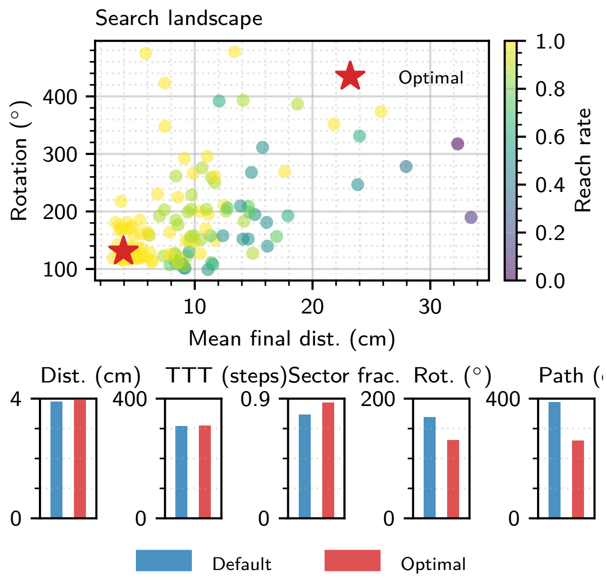

Figure 9: Two-phase TPE optimisation over eight controller parameters (150 trials). Left: per-trial objective score (faint points) and running best (line). Centre: search landscape, each trial placed at its mean final distance and sector fraction, coloured by reach rate, with the selected configuration starred. Right: default versus optimised controller across four metrics on the full evaluation grid (24 targets, 3×8 polar, $r \in [20, 55]$ cm). Source

Results

The selected configuration and the range searched for each parameter are given in Table 2; no optimum sits on a search bound. The most optimal controller's metrics are summarised in Table 3, its losses over time are plotted in Figure 10, an example trajectory is illustrated in Figure 11, and a heatmap of targets and the final distances to them is drawn in Figure 12.

The shoulder bands were scaled up and elbow bands were scaled down: the optimiser independently rediscovered the cross-talk hypothesis, that the elbow is "loud", driving the shoulder with it, so the controller converges on driving the elbow slowly. In addition, the minimum and maximum phase lengths are short, just a little larger than the state held by BMI, also rediscovering the hypothesis of small elbow movements that require quick and often shoulder corrections to minimise drastic cross-talk effects. The need for shoulder velocity damping is consolidated by the optimal controller converging on a high damping coefficient. The need for the elbow to be held once correct is also confirmed, due to the hard holding coefficient. The benefit of opposite direction shoulder drive in an effort to minimise cross-talk effects didn't seem to be valuable, likely due to the lack of granular control over selected muscle power. The most surprising result is that open-loop control wins over closed-loop control. We argue that the population-mean spectral target is only a loose proxy for the muscle selection latent state. In addition, as the `Brain` is a linear system, sinusoidal inputs produce sinusoidal outputs so a BMI based on rhythm can continue to operate without closed-loop control.

Parameter	Range	Optimal
Shoulder band gain	[0.50, 2.00]	1.5
Elbow band gain	[0.10, 1.50]	0.24
Cross-talk pre-drive	[0.05, 0.50]	0.14
Elbow hold once placed	[0.10, 0.80]	0.62
Shoulder velocity damping	[2.0, 50.0]	30.3
Min. phase length	[10, 60]	30
Max. phase length	[50, 200]	78
Close loop on pop. mean	{F, T}	F

Table 2: Optimal Phase-B controller hyperparameters from the TPE search (150 trials), shown with each parameter's search range. Source

Metric	Default (mean $\pm$ SD)	Optimal (mean $\pm$ SD)
Final dist. (cm)	3.90 ± 2.76	3.95 ± 1.42
Reach rate	1.000 ± 0.000	1.000 ± 0.000
TTT (steps)	308 ± 155	309 ± 147
Sector frac.	0.782 ± 0.126	0.871 ± 0.082
Rot. dist. (°)	169.3 ± 121.8	130.6 ± 65.5
Path length (cm)	389.4 ± 211.3	259.6 ± 68.4

Table 3: Mean $\pm$ SD of each metric across the 24-target grid for the default and Bayesian-optimised controllers (TTT, time to threshold). Source

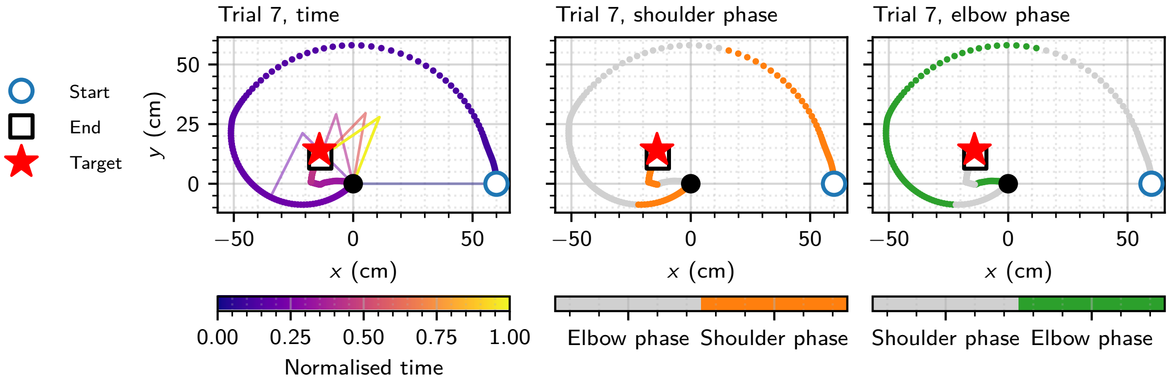

Figure 11: Hand trajectory for demo trial 7 (seed 0, $T = 1000$ steps). Left: path coloured by time, with 6 arm-posture snapshots at evenly spaced steps. Centre and right: the same path coloured by control phase, highlighting the shoulder- and elbow-correction phases respectively. Final distance to target 3.2 cm. Source

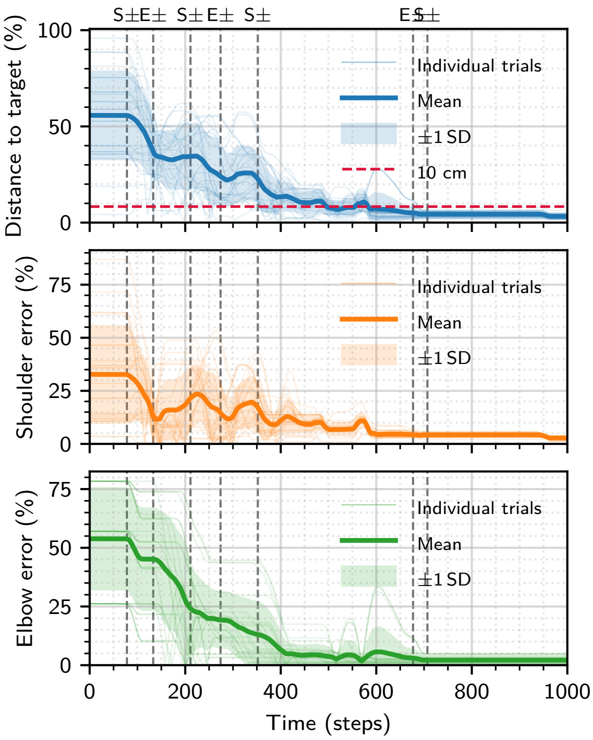

Figure 10: Error time series for the sequential open-loop controller on the 24-target grid (seed 0, $T = 1000$ steps). Top: distance to target as a percentage of the 120 cm maximum reach. Centre and bottom: shoulder and elbow joint error as a percentage of the 180° maximum wrapped error. Vertical dashed lines mark phase switches in demo trial 7, labelled by the activated phase (S± shoulder, E± elbow). 24/24 trials reached the 10 cm threshold (mean 309 steps); mean final distance 4.0 cm. Source

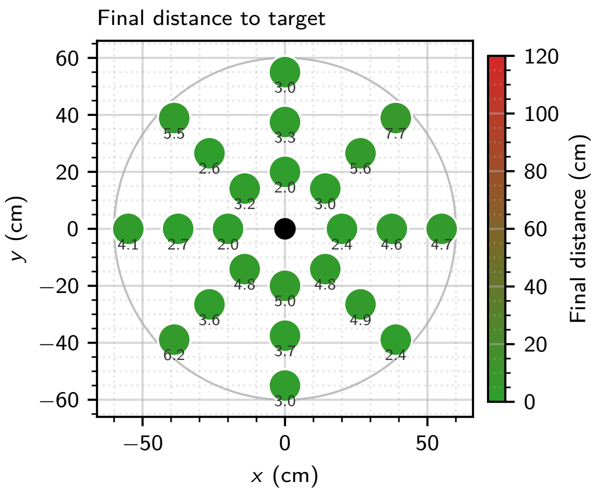

Figure 12: Final distance to target at each location of the 24-target grid (seed 0). The faint circle marks the arm reach boundary ($2L$, twice the link length). Source

Sensitivity to model parameters

We carry out ablation analysis, varying one parameter of the controller in isolation and holding everything else fixed to measure how sensitive performance is to that parameter. Hyperparameters under analysis include: the elbow band scale, how much the elbow gain is turned down relative to the shoulder gain Figure 13; the blend factor, how much we compensate for cross-talk by driving the shoulder in the opposite direction; closed-loop vs open-loop control Figure 14; and whether we constrain the elbow to a maximum of 180° or not Figure 15. We filter the metrics using a one-way Analysis of Variance (ANOVA), run independently for each metric across all conditions, with a null hypothesis that all group means are equal. A metric is plotted only if the ANOVA p-value satisfies $p \leq 0.05$; cross-talk compensation produced no significant figures.

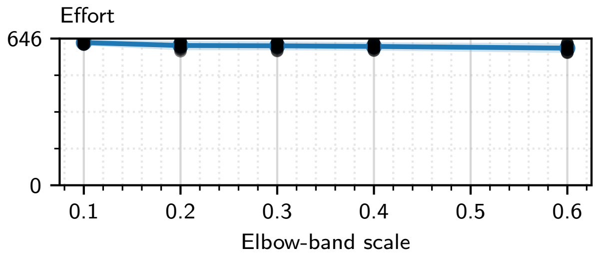

Figure 13: Ablation over the E± band-scale gain $\in \{0.1, 0.2, 0.3, 0.4, 0.6\}$ (S± fixed at 1.0) on 24 targets across $N = 10$ Brain seeds ($T = 1000$ steps). Source

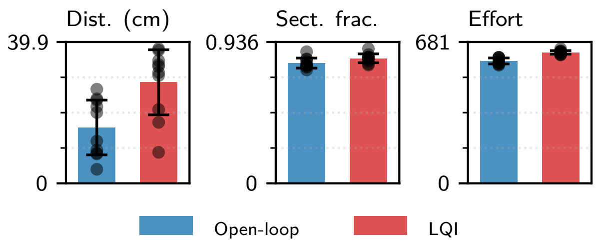

Figure 14: Open-loop (sinusoidal drive) versus closed-loop LQI control on 24 targets across $N = 10$ Brain seeds. Source

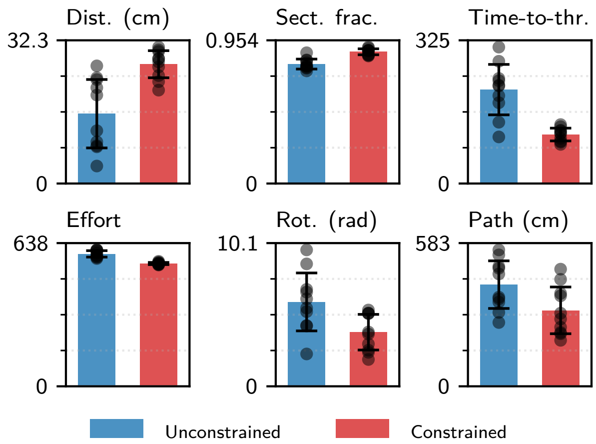

Figure 15: Unconstrained versus anatomically constrained elbow ($\theta_{el} \in [0, \pi]$). Source

Only the effort significantly varied with the elbow band scale (Figure 13). We find that open-loop control reduced the final distance, had less control effort, and slightly improved the sector fraction Figure 14. We find constraining the elbow to not extend past 180° worsens final distance, improves sector fraction, improves time-to-threshold, reduces effort, reduces rotation and improves the path. The constrained elbow is a much better controller in the areas reachable by an anatomically correct quadrant. However, in regions hard to access by such an arm (i.e., behind and towards the centre of the body, or negative-negative quadrant), its performance is poor, just like a real human. This is visualised in the heatmap in Figure 16, where if the human were imagined have their right shoulder joint at the origin, the poor final distances are behind the body, where a physiological elbow cannot reach.

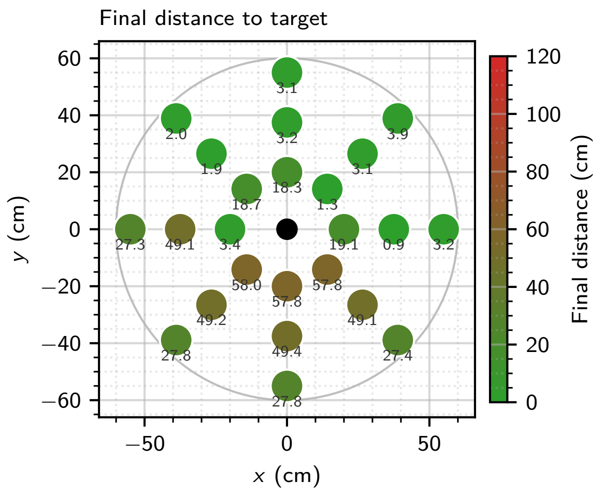

Figure 16: Final distance to target at each location of the 24-target grid (seed 0), for the controller optimised with the elbow anatomically constrained ($\theta_{el} \in [0, \pi]$), via an independent TPE search (150 trials). The faint circle marks the arm reach boundary ($2L$, twice the link length). Source

Robustness to noise

We evaluate control metrics after fitting to 1 seed only and applying it to 5 and compare to re-fitting each seed before application in Figure 17.

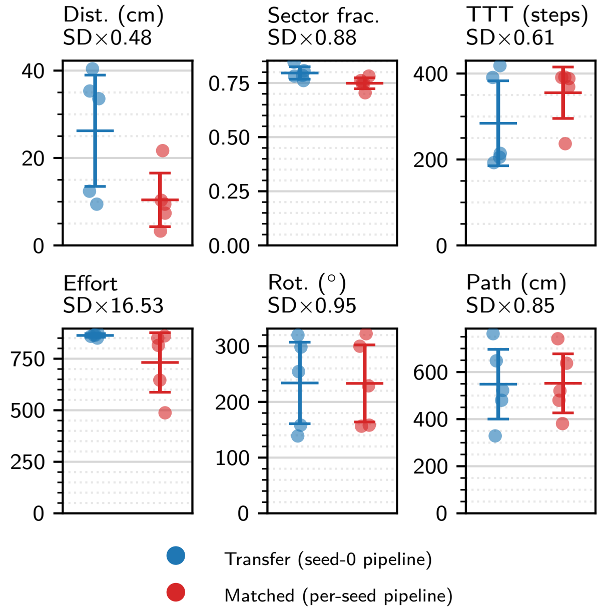

Figure 17: Per-seed pipeline robustness ($n = 5$ Brain seeds), one panel per metric. *Transfer*: the seed-0 pipeline evaluated on every seed. *Matched*: each seed's own end-to-end pipeline (ERA+EM $n_x = 4$, bands, closed-loop LQI gains, 80-trial TPE), evaluated on the 24-target grid. Points are per-seed means, bars mean ± SD; annotations give the SD ratio (Matched/Transfer, <1 means tighter). Source

We find that control metrics are highly seed-to-seed variable and perfom poorly until they are re-fit.

Alternative group approaches

One group memeber developed a control strategy with 2 levels of control, coarse and fine, with gains dynamically changing not when the elbow angle was correct, but rather the hand arrived within a specific range of the target. Their preliminary results lacked the small final distances of our method, but had significantly faster time to threshold and lower control effort. Another group member had a model-free approach that our hyperparameter optimisation converged on; their preliminary results also had competitive metrics which showed potential which inspired the inclusion of the boolean flag to toggle open- vs. closed-loop control in the optimiser.

Discussion

The strengths of our controller include: its reliability, a 100% reach rate across the target grid; its principled interpretability, as the optimiser independently rediscovered intuitive strategies; an honest methodology where only 1 seed is sampled throughout keeping the controller physically realisable; and high trajectory quality with a high sector fraction, low rotation, and small path length.

Limitations begin with a lack of evidence honestly recovering the true $n_x = 6$, and the difficulty in reconstructing the frequency response of u_2 to the same degree as u_1. The largest trade-off in the control strategy is controller interpretabilty, trajectory feasibility, and distance for the price of a poor time-to-threshold. This trade-off is established by the sequential focus on shoulder or elbow control, and limited attempts to control both joints at a time. In addition, the velocity damping to remove shoulder oscillations slows response time further. This decision was intuitively made on the basis of a shared power drive across selected muscles. Hyperparameter optimisation did not improve the headline final distance metric, it only made trajectories more feasible, highlighting this tradeoff further. We also hand-pick the scoring methods and each component's weights. This trade-off is mnaageable, as it is more convenient to have a predictable, more accurate prosthetic that is slow over an erratic, trajectory-inefficient one.

Conclusion

We identified the `Brain` as a 4-state LGSSM with a single-run ERA+EM pipeline, controlled its observation spectrum with an LQI controller, and used this to drive a sequential, cross-talk-aware hand controller tuned by TPE. The result reached every target on the grid within the success threshold, at a mean final distance under 4 cm. The design was not imposed by hand: the optimiser recovered the elbow-to-shoulder cross-talk asymmetry on its own, down-weighting elbow drive sixfold. The clearest result against expectation was that open-loop feedforward outperformed closing the loop on the population mean, which we attribute to that mean being a loose proxy for the muscle-selection latent and to the `Brain`'s linearity preserving rhythm without feedback. This accuracy and trajectory feasibility came at the cost of time-to-threshold, a direct consequence of driving one joint at a time.

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Code availability

Source code for analysis, plotting, and report typesetting can be found at: www.github.com/tahmidazam/gg4.